

Sanghee Kang, M.D., Ph.D.

Curriculum Vitae (Last updated: Aug/2026)

Professor · Assistant Dean for International Affairs, Korea University College of Medicine

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Contact Information

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Education

03/1998 – 02/2004 Korea University College of Medicine — Doctor of Medicine

09/2005 – 02/2008 Korea University Graduate School — Master of Medical Science

03/2009 – 02/2015 Korea University Graduate School — Ph.D. in Medical Science

Hospital / Academic Appointments

03/2004 – 02/2009 Korea University Ansan Hospital, Intern/Resident

05/2012 – 02/2014 Korea University Guro Hospital, Clinical Instructor

03/2014 – 02/2019 Korea University Guro Hospital, Clinical Assistant Professor

09/2016 – 08/2017 Dept. of Systems Biology, MD Anderson Cancer Center, Visiting Assistant Professor

03/2019 – 02/2023 Korea University Guro Hospital, Associate Professor

03/2023 – Present Korea University Guro Hospital, Professor

Professional Memberships and Roles

- General Secretary, Korean Hernia Society
- Director of Information, Korean Association of Robotic Surgeons
- Board Member, Korean BioChip Society (Secretary, In Vitro Diagnostic Devices Division)
- Editorial Board Member / Public Relations Committee Member, Korean Society of Coloproctology
- Education Committee Member, Korean Society of Surgical Oncology
- Academic Committee Member, Korean Surgical Skills Study Group

Ongoing Research Grants

11/2024 – 02/2029

Korean ARPA-H Project (KHIDI) — Development of a Microphysiological System for Safety Assessment of Immunotherapeutics. Subproject PI.

03/2026 – 02/2028

Safety Evaluation Research Program (MFDS) — Development of Predictive Evaluation Technology for the Immunotoxicity of Antibody Therapeutics. Subproject PI.

04/2026 – 02/2031

KIST Institutional Research Program (MSIT) — Clinical Concordance Evaluation Based on Patient-Derived Biomimetic Models. Subproject PI.

Publications (*first author, †corresponding author)

- Yim SY, Kim H, Kim TH, Kang SH, Lee Y, Choi E, et al. Identifying Sorafenib Benefit among Hepatocellular Carcinoma Patients: A Transcriptomic and Genomic Approach. *JHEP Reports*. 2026;101742.
- Cheong C, Doo HM, Ju Y, Bong JW, Kang SH, Lee SI, et al. Polyglycolic acid-cyanoacrylate complex for prevention of major intestinal anastomotic leakage in a rat model. *Ann Surg Treat Res*. 2025;109(5):335-343.
- Kim DY, Kim BG, Yun HM, Kim OH, Kang S, Bong JW, et al. Trans cinnamaldehyde enhances TRAIL induced apoptosis through ER stress mediated upregulation of DR5 in colorectal cancer cells. *Sci Rep*. 2025;15(1):38840.
- Bong JW, Lee H, Jeong S, Kang S†. Older Age Threshold for Oxaliplatin Benefit in Stage II to III Colorectal Cancer. *JAMA Netw Open*. 2025;8(8):e2525660.
- Ju Y, Kim BG, Gim JA, Bong JW, Cheong CO, Oh SC, et al. Synergistic Anticancer Effects of Fibroblast Growth Factor Receptor Inhibitor and Cannabidiol in Colorectal Cancer. *Nutrients*. 2025;17(16):2609.
- Kim KE, Bae SU, Lee SH, Lim DR, Ha HK, Kim J, et al. The articulated laparoscopic total mesorectal excision for rectal cancer (ATOME trial). *Trials*. 2025;26(1):260.
- Cho N, Jeong I, Ahn S, Gil H, Kim Y, Park J, Kang S, Lee H. Machine Learning to Assist in Managing Acute Kidney Injury in General Wards. *J Med Internet Res*. 2025;27:e66568.
- Ju Y, Bong JW, Cheong C, Kang S, Min BW, Lee SI. Effective utilization of polypectomy in endoscopic salvage treatment of rectal neuroendocrine tumors. *Ann Surg Treat Res*. 2024;107(3):151-157.
- Yang SS, Kye BH, Kang SH, Kim CH, Kim JH, Kim WR, Lee KY, Park IK. Intracorporeal anastomosis in minimally invasive right hemicolectomy: a nationwide survey. *Ann Surg Treat Res*. 2024;107(2):59-67.
- Bong JW, Kim JY, Ju Y, Cheong C, Kang S, Lee SI, Min BW. Optimal withdrawal time in initial surveillance colonoscopy after colorectal cancer surgery. *Ann Surg Treat Res*. 2024;107(4):212-220.
- Kim BG, Kim BR, Kim DY, Kim WY, Kang S, Lee SI, Oh SC. Cannabidiol enhances Atezolizumab efficacy by upregulating PD-L1 expression via the cGAS–STING pathway in triple-negative breast cancer cells. *Cancer Immunol Res*. 2024;12(12):1796-1807.
- Kim BR, Kim DY, Tran NL, Kim BG, Lee SI, Kang SH, et al. Daunorubicin induces GLI1-dependent apoptosis in colorectal cancer cell lines. *Int J Oncol*. 2024;64(6):1-13.
- Cho NJ, Jeong I, Kim Y, Kim DO, Ahn SJ, Kang SH, et al. A machine learning-based approach for predicting renal function recovery in general ward patients with acute kidney injury. *Kidney Res Clin Pract*. 2024;43(4):538.
- Bong JW, Kang S*, Park P. Multicenter study of prognostic factors in paraaortic lymph node dissection for metastatic colorectal cancer. *Ann Surg Treat Res*. 2023;105(5):271.
- Bong JW, Na Y, Ju Y, Cheong C, Kang S, Lee SI, Min BW. Nomogram for Predicting the Overall Survival of Underweight Patients with Colorectal Cancer. *BMC Gastroenterol*. 2023;23(1):39.

- Jeong YS, Eun YG, Lee SH, Kang SH*, Yim SY, Kim EH, Noh JK, et al. Clinically Conserved Genomic Subtypes of Gastric Adenocarcinoma. *Mol Cancer*. 2023;22(1):147.
- Kang AR, Kim JL, Kim YH, Kang S, Oh SC, Park JK. A Novel RIP1-Mediated Canonical Wnt Signaling Pathway That Promotes Colorectal Cancer Metastasis via β -Catenin Stabilization-Induced EMT. *Cancer Gene Ther*. 2023.
- Yoon S, Ji WB, Kim JS, Hong KD, Um JW, Min BW, Lee SI, et al. Long-Term Oncologic Outcome of D3 Lymph Node Dissection for Clinical Stage 2/3 Right-Sided Colon Cancer. *Int J Colorectal Dis*. 2023;38(1):42.
- Bong JW, Lee JA, Ju Y, Seo J, Kang SH, Lee SI, Min BW. Treatment Outcomes of Patients with Involved Resection Margin after Rectal Cancer Surgery. *Asia Pac J Clin Oncol*. 2022;18(4):378-87.
- Bong JW, Gim JA, Ju Y, Cheong C, Lee SI, Oh SC, Min BW, Kang S†. Prognosis and Sensitivity of Adjuvant Chemotherapy in Mucinous Colorectal Adenocarcinoma without Distant Metastasis. *Cancers*. 2022;14(5):1297.
- Hong KD, Hyun K, Um JW, Yoon SG, Shin J, Lee D, Baek SJ, et al. Clinical Outcomes of Surgical Management for Recurrent Rectal Prolapse. *Ann Surg Treat Res*. 2022;102(4):234-40.
- Lee SH, Yim SY, Jeong YS, Li QX, Kang SH*, Sohn BH, Kumar SV, et al. Consensus Subtypes of Hepatocellular Carcinoma Associated with Clinical Outcomes and Genomic Phenotypes. *Hepatology*. 2022;76(6):1634-48.
- Park S, Bong JW, Park I, Lee H, Choi J, Park P, Kim Y, Choi HS, Kang S†. Conbert: A Concatenation of Bidirectional Transformers for Standardization of Operative Reports from EMR. *Appl Sci*. 2022;12(21):11250.
- Kang S. Carcinoembryonic Antigen, the Most Accessible Test for Predicting Colorectal Cancer Prognosis. *Ann Coloproctol*. 2021;37(3):129-30.
- Kim NJ, Gim JA, Lee BJ, Choi BI, Park SB, Yoon HS, Kang SH, et al. RNA-Sequencing Identification and Validation of Genes Differentially Expressed in High-Risk Adenoma, Advanced Colorectal Cancer, and Normal Controls. *Funct Integr Genomics*. 2021;21:513-21.
- Kim NJ, Kim SM, Lee BJ, Yoon HS, Kang SH, Kim SH, Joo MK, Park JJ, Kim C. Detection of Microsatellite Instability in Colorectal Cancer Patients with a Plasma-Based Real-Time PCR Analysis. *Front Pharmacol*. 2021;12:758830.
- Kim JS, Baek SJ, Kwak JM, Kim J, Kim SH, Ji WB, Kim JS, et al. Impact of D3 Lymph Node Dissection on Upstaging and Short-Term Survival in Clinical Stage I Right-Sided Colon Cancer. *Asian J Surg*. 2021;44(10):1278-82.
- Kang S, Lee SI, Min BW, Lee TH, Baek SJ, Kwak JM, Kim J, et al. A Multicentre Comparative Study between Laparoscopic and Open Surgery for Intussusception in Adults. *Colorectal Dis*. 2020;22(10):1415-21.
- Yim SY, Kang SH*, Shin JH, Jeong YS, Sohn BH, Um SH, Lee JS. Low ARID1A Expression Is Associated with Poor Prognosis in Hepatocellular Carcinoma. *Cells*. 2020;9:2002.
- Kim HJ, Kang S*, Kim DY, You S, Park D, Oh SC, Lee DH. Diallyl Disulfide (DADS) Boosts TRAIL-Mediated Apoptosis in Colorectal Cancer Cells by Inhibiting Bcl-2. *Food Chem Toxicol*. 2019;125:354-60.
- Kang S, Kim EH, Hwang JE, Shin JH, Jeong YS, Yim SY, Joo EW, et al. Prognostic Significance of High Metabolic Activity in Breast Cancer: PET Signature in Breast Cancer. *Biochem Biophys Res Commun*. 2019;511(1):185-91.
- Kang S, Boo YJ, Lee JS, Han HJ, Jung CW, Kim CS. High Occupational Stress and Low Career Satisfaction of Korean Surgeons. *J Korean Med Sci*. 2014;30.
- Hong KD, Kang S, Kalaskar S, Wexner SD. Ligation of Intersphincteric Fistula Tract (LIFT) to Treat Anal Fistula: Systematic Review and Meta-Analysis. *Tech Coloproctol*. 2014;18(8):685-91.

Complete list of 40+ peer-reviewed publications available upon request; representative selection shown above.